Math 371 Exam 1 87/100 Name Darrell Long Page 1

(25 points)

1. a) Put the following formula into principle <u>conjunctive</u> normal form:

$(P \uparrow Q) \wedge (P \rightarrow (\neg Q \wedge S))$

$\neg(P \wedge Q) \wedge (P \rightarrow (\neg Q \wedge S))$ Def^n ($\uparrow$)

$\neg(P \wedge Q) \wedge (\neg P \vee (\neg Q \wedge S))$ E_{16}

$(\neg P \vee \neg Q) \wedge (\neg P \vee (\neg Q \wedge S))$ E_8

$(\neg P \vee \neg Q) \wedge (\neg P \vee \neg Q) \wedge (\neg P \vee S)$ E_7

$(\neg P \vee \neg Q) \wedge (\neg P \vee S)$ E_{10}

$(\neg P \vee \neg Q \vee S) \wedge (\neg P \vee \neg Q \vee \neg S) \wedge (\neg P \vee Q \vee S) \wedge (\neg P \vee \neg Q \vee S)$; $P \Leftrightarrow (P \vee Q) \wedge (P \vee \neg Q)$

$\boxed{(\neg P \vee Q \vee S) \wedge (\neg P \vee \neg Q \vee S) \wedge (\neg P \vee \neg Q \vee \neg S)}$

b) Use a) to write the formula in Π notation: Π_{467}

c) Use b) to write the formula in Σ notation: Σ_{01235}

d) Use c) to write the formula in principle <u>disjunctive</u> normal form:

$(P \wedge \neg Q \wedge S) \vee (\neg P \wedge Q \wedge S) \vee (\neg P \wedge Q \wedge \neg S) \vee (\neg P \wedge \neg P \wedge S) \vee (\neg P \wedge \neg Q \wedge \neg S)$

(20 points)

2. The connective * is given by the truth table at the right. Support your answers to the following questions (you can use truth tables if you like).

P	Q	P*Q	$\neg P \wedge Q$
T	T	F	F
T	F	F	F
F	T	T	T
F	F	F	F

a) Can * be written solely in terms of $\downarrow$? (If so, give a formula.)

-2 $((P \uparrow P) \uparrow Q) \uparrow ((P \uparrow P) \uparrow Q)$

b) Is {*, $\neg$} functionally complete? Y

$\neg P \Leftrightarrow \neg P$

$P \wedge Q \Leftrightarrow \neg P * Q$

P	Q	$\neg P$	$\neg P * Q$	$P \wedge Q$
T	T	F	T	T
	F	F	F	F
T	T	T	F	F
F	F	T	F	F
F				

(5 points)

3. Give the Generalized DeMorgan's equivalent of the formula:

$(P \wedge \neg Q) \uparrow \neg(P \downarrow F)$

-1

$\neg((\neg P \vee Q) \downarrow \neg(\neg P \uparrow T))$ — leave not signs that exist alone.

(5 points)

4. Explain the difference between $\Rightarrow$ and $\rightarrow$.

$\rightarrow$ is an operator, like $\wedge, \vee, +, -$ etc.

$\Rightarrow$ says something is true $(A \rightarrow B)$.

(10 points)

5. Prove $\neg(P \rightleftarrows R) \Leftrightarrow (P \wedge \neg R) \vee (\neg P \wedge R)$ <u>without</u> using truth tables; identify each equivalence that you use.

$\neg(P \rightleftarrows R) \Leftrightarrow (P \wedge \neg R) \vee (\neg P \wedge R)$

$\Leftrightarrow \neg((P \rightarrow R) \wedge (R \rightarrow P))$ E_{21}

$\Leftrightarrow \neg((\neg P \vee R) \wedge (\neg R \vee P))$ E_{16}

$\Leftrightarrow \neg(\neg P \vee R) \vee \neg(\neg R \vee P)$ E_8

$\Leftrightarrow \neg(\neg P \vee R) \vee \neg(P \vee \neg R)$ E_3

$\Leftrightarrow (\neg\neg P \wedge \neg R) \vee (\neg P \wedge \neg\neg R)$ E_9

$\Leftrightarrow (P \wedge \neg R) \vee (\neg P \wedge R)$ E_1

QED

#5: $\neg(P \rightleftarrows R) \Leftrightarrow$

$P \rightleftarrows \neg R \Leftrightarrow$ E_{20}

$(P \wedge \neg R) \vee (\neg P \wedge \neg\neg R) \Leftrightarrow$ E_{22}

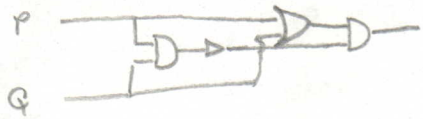
$(P \wedge \neg R) \vee (\neg P \wedge R) \Leftrightarrow$ E_1

#8: H_1: $\neg P \wedge Q$ H_2: $R \to P$ H_3: $\neg Q \vee R$

1)	$\neg P \wedge Q$	P	
2)	$\neg P$	T	I_1 (1)
3)	Q	T	I_2 (1)
4)	$R \to P$	P	
5)	$\neg Q \vee R$	P	
6)	$Q \to R$	T	E_{16} (5)
7)	$Q \to P$	T	I_{14} (6,4)
8)	$\neg Q \vee P$	T	E_{16} (7)
9)	$(\neg Q \vee P) \wedge \neg P$	T	I_9 (8,2)
10)	$(\neg P \wedge \neg Q) \vee (P \wedge \neg P)$	T	E_6 (9)
11)	$\neg P \wedge \neg Q$	T	$P \vee F \Leftrightarrow P$
12)	$\neg P \wedge \neg Q \wedge Q$	T	I_9 (11,3)
13)	F	T	E_{15} (12)

(5 points)

6. Draw a circuit diagram, using only AND, OR, and NOT gates, to implement the Exclusive-Or connective, $P \bar{\vee} Q$. (Hint: first find an appropriate formula equivalent to $P \bar{\vee} Q$.) $P \bar{\vee} Q \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$



(10 points)

7. Support your answers to the following (you may use truth tables if you like).

Does $R \uparrow Q \Rightarrow R \downarrow Q$? NO (back)

Does $R \downarrow Q \Rightarrow R \uparrow Q$? Yes (back)

(20 points)

8. Prove that the hypotheses H_1: $\neg P \wedge Q$, H_2: $R \rightarrow P$, and H_3: $\neg Q \vee R$ are inconsistent; identify each implication that you use.

-10

H_1: $\neg P \wedge Q$ H_2: $R \rightarrow P$ H_3: $\neg Q \vee R$; (need R & Q)

1)	$\neg P \wedge Q$	P
2)	$R \rightarrow P$	P
3)	$\neg Q \vee R$	P
4)	$Q \rightarrow R$	T E_{16}
5)	$Q \rightarrow P$	T I_{13} (4,2)
6)	Q	P (assumed)
7)	P	T I_{11} (5,6)
8)	$Q \wedge \neg P$	T E_2 (1)
9)	$\neg P$	T I_2 (8)
10)	$P \wedge \neg P$	T I_9 (7,9)

If I can do this

6)	$\neg R \vee P$	T E_{16} (2)
7)	$P \rightarrow Q$	T E_{16} (1)
8)	$Q \rightarrow P$	T (5)

F

1)	$\neg P \wedge Q$	P
2)	$\neg P$	T I_1
3)	Q	T I_2
4)	$R \rightarrow P$	P
5)	$\neg R$	I_{12} (2,4)
6)	$\neg Q \vee R$	P
7)	$R \vee \neg Q$	T E_3
8)	$\neg Q$	T
9)	$Q \wedge \neg Q$	T I_9 (3,8)

7) A: $R \uparrow Q \Rightarrow R \downarrow Q$

1)	$(R \uparrow Q)$	P
2)	$\neg(R \wedge Q)$	T Def^n
3)	$\neg R \vee \neg Q$	T E_3
4)	$\neg(R \downarrow Q)$	P assumed
5)	$\neg\neg(R \vee Q)$	T Def^n
6)	$R \vee Q$	T E_1

R	Q	$R \uparrow Q$	$R \downarrow Q$
T	T	F	F
T	F	T	F
F	T	T	F
F	F	T	T

C: $R \downarrow Q \Rightarrow R \uparrow Q$

1)	$R \downarrow Q$	P
2)	$\neg(R \vee Q)$	T Def^n
3)	$\neg R \wedge \neg Q$	T E_9
4)	$\neg R$	T I_1
5)	$\neg R \vee \neg Q$	T I_3
6)	$\neg(R \wedge Q)$	T E_8
7)	$R \uparrow Q$	T Def^n